How a factory makes chilled beef lasagna

A representative oven-cooked, rapidly chilled ready meal for reheating. Follow each column down, then continue at the matching letter.

Process / equipment · Decision: follow Yes / No · Hold / corrective action · A Matching letters connect the three process sections

1 Ingredients + preparation

01 Receive and inspect ingredients

Beef, vegetables, flour/semolina, milk, fat, cheese and packaging. Check approved supplier, identity, condition, temperature and lot codes.

Ingredients meet specification?

- No → **QUARANTINE** Investigate lot; reject if unacceptable.
- Yes

02 Store, segregate and clear the line

Keep beef chilled; segregate raw materials and allergens. Use oldest valid stock first. Verify cleaning, equipment setup, scales and the approved recipe.

Line and recipe cleared to start?

- No → **STOP / RESET** Clean or correct setup; repeat this check.
- Yes

03 Weigh and issue the recipe

Meter each ingredient; link quantities and lots to one production batch.

04a Beef ragù

In a jacketed kettle, brown minced beef and aromatics. Add tomato and herbs; simmer with agitation. Check cook record, batch yield and consistency.

04b Béchamel

Heat fat and flour for a roux; meter in milk. A scraped, stirred kettle heats and thickens the sauce. Check temperature and flow consistency.

04c Pasta

Mix flour/semolina and water (egg if specified). Sheet and calibrate thickness; precook, quench, drain and cut. Check size and texture.

Release conforming components; hold failures for QA. Control transfer time and temperature.

A → top of section 2

2 Assembly + cooking

A

05 Feed the trays

Denester separates food-grade, ovenable trays onto a conveyor.

06 Deposit and layer to the recipe

Piston/lobe-pump depositors meter base sauce; pick-and-place loads pasta sheets. Add measured ragù and béchamel. Keep edges and tray rims clear.

More recipe layers needed?

- Yes → 06 Deposit and layer to the recipe
- No

07 Finish and check the fill

Add final sauce and cheese. Check layer position, portion weight and clean sealing rim.

Layers, weight and rim correct?

- No → **DIVERT TRAY** Correct only under an approved procedure; recheck or reject.
- Yes

08 Cook in a continuous oven

Set zones, airflow and belt residence time for the validated product/load. Verify the coldest product location reaches the required core time–temperature.

Validated cook achieved?

- No → **HOLD LOT** QA authorises validated reprocessing or rejects the lot. Log deviation.
- Yes

B → section 3

B

3 Chill + pack + dispatch

B

09 Rapidly chill the cooked trays

Blast/spiral chiller removes heat. Control load, airflow and residence time; record the core cooling curve against the validated limit.

Cooling profile within limit?

- No → **HOLD LOT** Investigate fault. QA decides lot status. Rechilling cannot prove safety.
- Yes

10 Seal in a hygienically controlled area

Tray-sealer applies barrier film to clean rims. Check seal integrity. Any modified-atmosphere gas mix and shelf life require product-specific validation.

11 Inspect, identify and reconcile

Verify label/recipe match, allergens, use-by date, lot code, storage and reheating instructions. Check final weight and seals; use a validated foreign-body detector suited to the tray (e.g. X-ray). Reconcile batch and cook/chill records.

All release checks and records pass?

- No → **BLOCK RELEASE** Segregate affected lot. Investigate; correct only if authorised, then repeat failed checks.
- Yes

12 QA release and chilled dispatch

Authorised quality staff release the lot. Case-pack, palletise and track it into refrigerated storage and transport; monitor the specified cold chain.

RETAIL → CUSTOMER REHEATS TO LABEL

Controls to understand HACCP (Hazard Analysis and Critical Control Points) determines CCPs and limits. A common core lethality reference is 70 °C for 2 minutes or equivalent [6]; validate for the product/load. Chill rapidly to validated limits; a typical cold-chain target is ≤5 °C [6]. Separate raw/cooked areas; control wheat/gluten, milk and egg. Log lots, temperatures and actions. Shelf life needs microbiological and quality validation.

Route variants Some factories assemble chilled cooked components or cook in-pack. A frozen route uses validated freezing, freezer-compatible packs and a frozen cold chain. Freezing does not sterilise food.

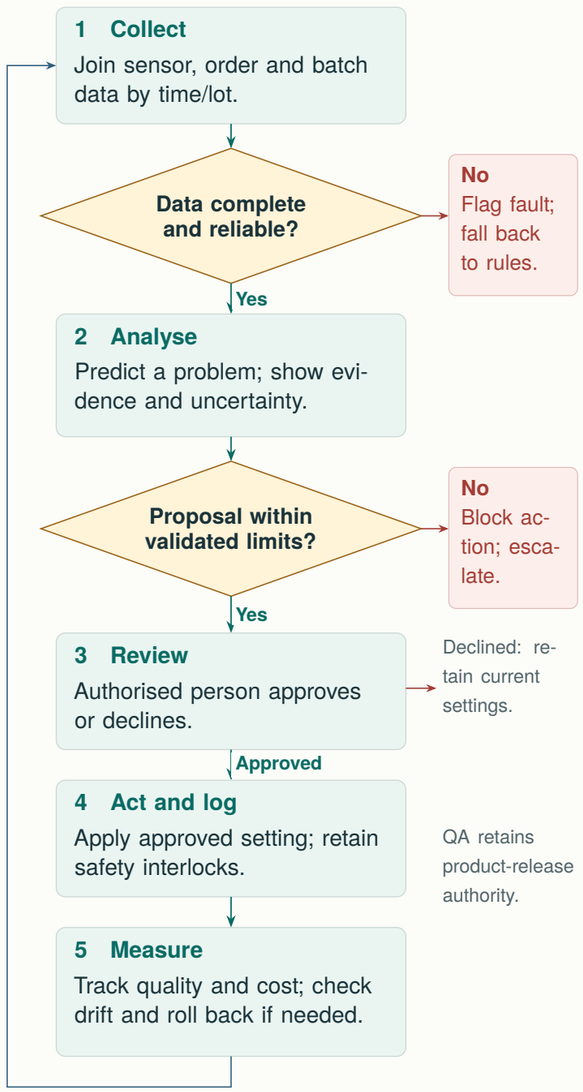
Sources / further reading: [1] Robot Food Tech: layering and oven feed [2] Italgi: pasta preparation [3] GS Italia: dosing and weight control [4] Fabbri: tray-sealing lines [5] Food Standards Agency: HACCP principles [6] Chilled Food Association: heat, cooling and packaging [7] CFA: shelf life

Where AI could improve the factory

A proposed decision-support tool for the same chilled-lasagna line. Benefits are hypotheses to test in a controlled pilot.

ML learns patterns. **Optimisation** selects feasible actions. **PLC automation** executes fixed rules. **LLM assistance** explains approved information.

THE DECISION LOOP



01 Forecast demand and schedule production

LINK: 01–03, 12 | FORECASTING ML + OPTIMISATION

Data: Orders, sales, stock life, line capacity, changeovers.

Analysis: Forecast demand; schedule feasible batches around material, allergen-cleaning and dispatch constraints.

Action: Planner approves batch quantities and sequence.

Measure: Forecast error, waste kg, on-time delivery, good trays/hour.

02 Stabilise sauce and portion consistency

LINK: 04, 06–07 | REGRESSION / ANOMALY DETECTION

Data: Sauce temperature, measured flow/viscosity, pump settings, ingredient lots and checkweigher trends.

Analysis: Predict dose drift and identify likely contributors.

Action: Recommend bounded adjustments; approve and verify against recipe and weight requirements.

Measure: Giveaway g/tray, weight SD, underweight rate, sauce consistency.

03 Inspect layers, seals and labels

LINK: 06–07, 10–11 | COMPUTER VISION + OCR

Data: Images, approved artwork and labelled defect examples.

Analysis: Detect missing/folded sheets, contaminated rims or label mismatch; read codes with OCR.

Action: Flag/divert suspect packs for validated inspection. Cameras cannot prove microbiological safety or seal integrity.

Measure: Missed defects, false rejects, review burden, complaint rate.

04 Improve oven and chiller performance

LINK: 08–09 | PREDICTIVE MODEL + OPTIMISATION

Data: Core profiles, zone/air temperatures, belt speed, load, ambient conditions and energy meters.

Analysis: Predict load effects and feasible energy savings.

Action: Propose load/settings changes within validated limits. Preserve independent cook/chill monitoring and alarms.

Measure: kWh/kg of good product, profile compliance, good trays/hour.

05 Anticipate equipment problems

LINK: 04–10 | CONDITION MONITORING ML

Data: Pump/sealer vibration, motor current, temperature, fault history, maintenance and operating conditions.

Analysis: Detect unusual wear patterns; estimate failure risk.

Action: Engineer verifies the alert and schedules inspection during a planned stoppage or changeover.

Measure: Unplanned downtime, warning lead time, false alarms, cost.

06 Trace lots and support operators

LINK: ALL STAGES | DATABASE QUERIES + GROUNDED LLM

Data: Lot genealogy, approved SOPs, recipes and alarm logs.

Analysis: Query affected lots; explain instructions with source references. Lot matching uses deterministic records.

Action: Present checks and evidence to operators/QA. Abstain when sources are missing; log advice and decisions.

Measure: Trace time, answer accuracy, task time, escalation rate.

System boundary: Connect read-only to planning, production and quality records first. Record model version, input time, recommendation, approval and outcome. Conventional PLC safety controls remain independent.

PILOT 1 Establish a baseline

Start with portion giveaway. Calibrate scales; join batches and settings. Measure normal variation across lots, shifts and recipes.

PILOT 2 Run in shadow mode

Test on later, unseen batches. Compare ML with current rules; review missing data, drift and false alarms before acting on advice.

PILOT 3 Trial approved actions

Use matched shifts/recipes and agreed success thresholds. Count waste, weight spread, inspection errors, downtime, energy and output.

PILOT 4 Expand or roll back

Scale only after measured benefit with no deterioration in safety or quality. Stop advice on drift, poor data or adverse results; investigate.

Evidence and definitions: [8] NIST: evaluate maintenance AI using algorithm and production metrics [9] NIST: combine measurement, physical models and AI

Applications above are a proposed design, not reported results from a lasagna factory. ML = machine learning; PLC = programmable logic controller; LLM = large language model; OCR = optical character recognition; QA = quality assurance; SD = standard deviation.